

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414677
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Miller; Scott

Endoscope accessory with locking elements

Abstract

An endoscope accessory, such as a coupler device, is provided that includes a main body with a visualization section for allowing viewing of tissue by the endoscope and a proximal end configured for attachment to a distal end portion of the endoscope. The device includes a passage within the main body having an open distal end and a proximal end aligned with the working channel of the endoscope. A locking element on the coupler device secures an angle of exit of an instrument passing through the passage.

Inventors:	Miller; Scott (Arlington, VA)
Applicant:	GI Scientific, LLC (Arlington, VA)
Family ID:	1000008818282
Assignee:	GI Scientific, LLC (Arlington, VA)
Appl. No.:	18/422229
Filed:	January 25, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240156332 A1	May. 16, 2024

Related U.S. Application Data

continuation parent-doc US 17572850 20220111 US 11910999 child-doc US 18422229
continuation parent-doc US 16897098 20200609 US 11253137 20220222 child-doc US 17572850
continuation parent-doc US 15746196 US 10856724 20201208 WO PCT/US2016/433721
20160721 child-doc US 16897098
us-provisional-application US 62195291 20150721

Publication Classification

Int. Cl.: **A61B1/00** (20060101); **A61B1/018** (20060101); **A61B1/06** (20060101); **A61B1/273** (20060101)

U.S. Cl.:

CPC **A61B1/00098** (20130101); **A61B1/00101** (20130101); **A61B1/00105** (20130101); **A61B1/00131** (20130101); **A61B1/00137** (20130101); **A61B1/00142** (20130101); **A61B1/00177** (20130101); **A61B1/018** (20130101); **A61B1/0676** (20130101); **A61B1/2736** (20130101); A61B1/00089 (20130101); A61B1/00103 (20130101); A61B1/00165 (20130101); A61B1/00181 (20130101)

Field of Classification Search

CPC: A61B (1/00098); A61B (1/00089); A61B (1/00133); A61B (1/00137); A61B (1/01); A61B (1/018)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3774614	12/1972	Cook	N/A	N/A
3858577	12/1974	Bass et al.	N/A	N/A
4090501	12/1977	Chaitin	N/A	N/A
4201199	12/1979	Smith	N/A	N/A
4207872	12/1979	Meiri et al.	N/A	N/A
4340811	12/1981	Yamashita et al.	N/A	N/A
4681093	12/1986	Ono et al.	N/A	N/A
4696544	12/1986	Costella	N/A	N/A
4744620	12/1987	Ueno et al.	N/A	N/A
4779611	12/1987	Grooters et al.	N/A	N/A
4805598	12/1988	Ueda	N/A	N/A
4878725	12/1988	Hessel et al.	N/A	N/A
4881810	12/1988	Hasegawa	N/A	N/A
4888243	12/1988	Jonas et al.	N/A	N/A
4967732	12/1989	Inoue	N/A	N/A
5040715	12/1990	Green et al.	N/A	N/A
5050585	12/1990	Takahashi	N/A	N/A
5080660	12/1991	Buelna	N/A	N/A
5104025	12/1991	Main et al.	N/A	N/A
5137198	12/1991	Nobis et al.	N/A	N/A
5201900	12/1992	Nardella	N/A	N/A
5205459	12/1992	Brinkerhoff et al.	N/A	N/A
5237984	12/1992	Williams, III et al.	N/A	N/A
5271379	12/1992	Phan et al.	N/A	N/A
5326013	12/1993	Green et al.	N/A	N/A
5329935	12/1993	Takahashi	N/A	N/A
5337734	12/1993	Saab	N/A	N/A

5342388	12/1993	Toller	N/A	N/A
5413052	12/1994	Breezer et al.	N/A	N/A
5413268	12/1994	Green et al.	N/A	N/A
5443781	12/1994	Saab	N/A	N/A
5448990	12/1994	De Faria-Correa	N/A	N/A
5460168	12/1994	Masubuchi et al.	N/A	N/A
5476206	12/1994	Green et al.	N/A	N/A
5536236	12/1995	Yabe et al.	N/A	N/A
5555129	12/1995	Konno et al.	N/A	N/A
5569157	12/1995	Nakazawa et al.	N/A	N/A
5575291	12/1995	Hayakawa et al.	N/A	N/A
5605532	12/1996	Schermerhorn	N/A	N/A
5632717	12/1996	Yoon	N/A	N/A
5657921	12/1996	Young et al.	N/A	N/A
5662258	12/1996	Knodel et al.	N/A	N/A
5674181	12/1996	Iida	N/A	N/A
5707342	12/1997	Tanaka	N/A	N/A
5725474	12/1997	Yasui et al.	N/A	N/A
5725475	12/1997	Yasui et al.	N/A	N/A
5738629	12/1997	Moll et al.	N/A	N/A
5743851	12/1997	Moll et al.	N/A	N/A
5771327	12/1997	Bar-Or et al.	N/A	N/A
5788628	12/1997	Matsuno et al.	N/A	N/A
5808813	12/1997	Lucey et al.	N/A	N/A
5840014	12/1997	Miyano et al.	N/A	N/A
5860913	12/1998	Yamaya et al.	N/A	N/A
5897487	12/1998	Ouchi	N/A	N/A
5916148	12/1998	Tsuyuki	N/A	N/A
6059719	12/1999	Yamamoto et al.	N/A	N/A
6131789	12/1999	Schulze et al.	N/A	N/A
6217509	12/2000	Foley et al.	N/A	N/A
6250532	12/2000	Green et al.	N/A	N/A
6277065	12/2000	Donofrio	N/A	N/A
6283951	12/2000	Flaherty et al.	N/A	N/A
6293907	12/2000	Axon et al.	N/A	N/A
6306081	12/2000	Ishikawa et al.	N/A	N/A
6409725	12/2001	Khandkar et al.	N/A	N/A
6416462	12/2001	Tovey et al.	N/A	N/A
6673091	12/2003	Shaffer et al.	N/A	N/A
6689130	12/2003	Arai et al.	N/A	N/A
6699180	12/2003	Kobayashi	N/A	N/A
6712524	12/2003	Beatty et al.	N/A	N/A
6723350	12/2003	Burrell et al.	N/A	N/A
6733440	12/2003	Ailinger et al.	N/A	N/A
6770069	12/2003	Hobart et al.	N/A	N/A
6792837	12/2003	Battistone	N/A	N/A
6855108	12/2004	Ishibiki et al.	N/A	N/A
6866627	12/2004	Nozue	N/A	N/A
6934093	12/2004	Kislev et al.	N/A	N/A
6981628	12/2005	Wales	N/A	N/A

6988650	12/2005	Schwemmer et al.	N/A	N/A
7033317	12/2005	Pruitt	N/A	N/A
7046439	12/2005	Kaminsky et al.	N/A	N/A
7087012	12/2005	Ishibiki	N/A	N/A
7112195	12/2005	Boll et al.	N/A	N/A
7205339	12/2006	Muratoglu	N/A	N/A
7235592	12/2006	Muratoglu et al.	N/A	N/A
7238153	12/2006	Moriyama	N/A	N/A
7245813	12/2006	Brown et al.	N/A	N/A
7464847	12/2007	Viola et al.	N/A	N/A
7537561	12/2008	Yamaya et al.	N/A	N/A
7553278	12/2008	Kucklick	N/A	N/A
7554743	12/2008	Jiang et al.	N/A	N/A
7566993	12/2008	May	N/A	N/A
7621868	12/2008	Breidenthal et al.	N/A	N/A
7819872	12/2009	Johnson et al.	N/A	N/A
7977255	12/2010	Scheer et al.	N/A	N/A
8180423	12/2011	Mang et al.	N/A	N/A
8241204	12/2011	Spivey	N/A	N/A
8905921	12/2013	Titus	N/A	N/A
8915931	12/2013	Boudreaux et al.	N/A	N/A
9459442	12/2015	Miller	N/A	N/A
9709795	12/2016	Miller	N/A	N/A
10101574	12/2017	Miller	N/A	N/A
2002/0035311	12/2001	Ouchi	N/A	N/A
2002/0065515	12/2001	Falwell et al.	N/A	N/A
2002/0133148	12/2001	Daniel et al.	N/A	N/A
2003/0040657	12/2002	Yamaya et al.	N/A	N/A
2003/0181900	12/2002	Long	N/A	N/A
2004/0157073	12/2003	Burrell et al.	N/A	N/A
2004/0249246	12/2003	Campos	N/A	N/A
2004/0263613	12/2003	Morita	N/A	N/A
2004/0267092	12/2003	Ishibiki	N/A	N/A
2005/0043589	12/2004	Pruitt	N/A	N/A
2005/0080411	12/2004	Ouchi	N/A	N/A
2005/0131279	12/2004	Boulais et al.	N/A	N/A
2005/0197530	12/2004	Wallace et al.	N/A	N/A
2005/0234297	12/2004	Devierre et al.	N/A	N/A
2006/0030844	12/2005	Knight et al.	N/A	N/A
2006/0084839	12/2005	Mourlas et al.	N/A	N/A
2006/0173241	12/2005	Ouchi et al.	N/A	N/A
2006/0200176	12/2005	Matsuno et al.	N/A	N/A
2006/0229662	12/2005	Finkielsztejn et al.	N/A	N/A
2006/0270900	12/2005	Chin et al.	N/A	N/A
2006/0270978	12/2005	Binmoeller	N/A	N/A
2007/0038043	12/2006	Gelikonov et al.	N/A	N/A
2007/0066869	12/2006	Hoffman	N/A	N/A
2007/0066870	12/2006	Ohashi et al.	N/A	N/A
2007/0073108	12/2006	Takahashi	N/A	N/A
2007/0208219	12/2006	Carter	N/A	N/A

2007/0239620	12/2006	Schwartz et al.	N/A	N/A
2007/0249898	12/2006	Otawara	N/A	N/A
2007/0260117	12/2006	Zwolinski et al.	N/A	N/A
2007/0270897	12/2006	Skerven	N/A	N/A
2007/0282256	12/2006	Hu et al.	N/A	N/A
2007/0293888	12/2006	Harren et al.	N/A	N/A
2008/0021268	12/2007	Shoroji et al.	N/A	N/A
2008/0021269	12/2007	Tinkham et al.	N/A	N/A
2008/0033246	12/2007	Matsui et al.	N/A	N/A
2008/0139885	12/2007	Knapp	N/A	N/A
2008/0177135	12/2007	Muyari et al.	N/A	N/A
2008/0188874	12/2007	Henderson	N/A	N/A
2008/0262295	12/2007	Kendale et al.	N/A	N/A
2008/0306335	12/2007	Lau et al.	N/A	N/A
2009/0048483	12/2008	Yamamoto	N/A	N/A
2009/0048486	12/2008	Surti	N/A	N/A
2009/0062790	12/2008	Malchano et al.	N/A	N/A
2009/0098409	12/2008	Yamada et al.	N/A	N/A
2009/0124858	12/2008	Oskin et al.	N/A	N/A
2009/0143643	12/2008	Weitzner et al.	N/A	N/A
2009/0156898	12/2008	Ichimura	N/A	N/A
2009/0254164	12/2008	Johnson et al.	N/A	N/A
2009/0315989	12/2008	Adelson	N/A	N/A
2009/0326328	12/2008	Kucklick	N/A	N/A
2010/0026940	12/2009	Takegami et al.	N/A	N/A
2010/0121442	12/2009	Shea et al.	N/A	N/A
2010/0017414	12/2009	Hsu	N/A	N/A
2010/0203454	12/2009	Brongersma et al.	N/A	N/A
2010/0268027	12/2009	Aono et al.	N/A	N/A
2010/0286475	12/2009	Robertson	N/A	N/A
2011/0124960	12/2010	St. Onge et al.	N/A	N/A
2011/0152618	12/2010	Surti	N/A	N/A
2011/0241460	12/2010	Jiang	N/A	N/A
2012/0034573	12/2011	Erdmann et al.	N/A	N/A
2012/0209074	12/2011	Titus	N/A	N/A
2012/0209090	12/2011	Goodall et al.	N/A	N/A
2012/0232342	12/2011	Reydel	N/A	N/A
2013/0040516	12/2012	Pruneri et al.	N/A	N/A
2013/0090527	12/2012	Axon	N/A	N/A
2013/0144287	12/2012	Crowley et al.	N/A	N/A
2013/0190562	12/2012	Smith et al.	N/A	N/A
2013/0237998	12/2012	Wallace et al.	N/A	N/A
2014/0027578	12/2013	Comtesse	N/A	N/A
2015/0065795	12/2014	Titus	N/A	N/A
2015/0073214	12/2014	Ueda	N/A	N/A
2016/0029875	12/2015	Okada	N/A	N/A
2016/0051135	12/2015	Greenberg et al.	N/A	N/A
2016/0270636	12/2015	Iwasaka et al.	N/A	N/A
2017/0066111	12/2016	Wang	N/A	N/A
2017/0251910	12/2016	Surti et al.	N/A	N/A

2017/0311789	12/2016	Mulcahey et al.	N/A	N/A
2017/0311974	12/2016	Friedrichs	N/A	N/A
2018/0160886	12/2017	Govani et al.	N/A	N/A
2018/0206708	12/2017	Miller	N/A	N/A
2019/0009963	12/2018	Wessely	N/A	N/A
2019/0070998	12/2018	Spencer et al.	N/A	N/A
2020/0100655	12/2019	Morishima et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1692872	12/2010	CN	N/A
3532609	12/1986	DE	N/A
1870018	12/2006	EP	N/A
H0373168	12/1990	JP	N/A
07178094	12/1994	JP	N/A
H09238893	12/1996	JP	N/A
05123288	12/1998	JP	N/A
2000300570	12/1999	JP	N/A
3124079	12/2000	JP	N/A
200217655	12/2001	JP	N/A
2002233491	12/2001	JP	N/A
2003033319	12/2002	JP	N/A
2003339631	12/2002	JP	N/A
2005066139	12/2004	JP	N/A
2006026344	12/2005	JP	N/A
2008029384	12/2007	JP	N/A
2009261830	12/2008	JP	N/A
2010063721	12/2009	JP	N/A
WO 9929362	12/1998	WO	N/A
WO 2001085319	12/2000	WO	N/A
WO 2006138409	12/2005	WO	N/A
WO 2007029230	12/2006	WO	N/A
WO 2007029814	12/2006	WO	N/A
WO 2007147060	12/2006	WO	N/A
WO 2009149042	12/2008	WO	N/A
WO 2011085319	12/2010	WO	N/A
WO 2011099329	12/2010	WO	N/A
WO 2011148172	12/2010	WO	N/A
WO 2014123563	12/2013	WO	N/A
WO 2017011535	12/2016	WO	N/A

OTHER PUBLICATIONS

European Examination Report for EP Appl. No. 15843356.5 dated May 20, 2019, 7 pages. cited by applicant

Extended European Search Report for EP Appl. No. 16828548.4 dated Feb. 28, 2019. cited by applicant

Extended European Search Report for EP Appl. No. 16804476.6 dated Dec. 5, 2018. cited by applicant

Extended European Search Report and Written Opinion for EP Appl. No. 16804462-6 dated Dec. 10, 2018. cited by applicant

Extended European Search Report for EP Appl. No. 12747511.9 dated Jan. 3, 2018. cited by applicant

Extended European Search Report for EP Appl. No. 18174913.6 dated Aug. 16, 2019. cited by applicant

International Preliminary Report on Patentability issued in PCT/US2015/051662 dated Apr. 6, 2017. cited by applicant

International Search Report and Written Opinion for PCT Appl. No. PCT/US2019/012448 mailed Apr. 16, 2019. cited by applicant

International Search Report and Written Opinion dated Oct. 26, 2016 for PCT Application No. PCT/US2016/043371, filed Jul. 21, 2016. cited by applicant

International Search Report issued in corresponding International Application No. PCT/US2015/051662 dated Dec. 14, 2015. cited by applicant

International Search Report and Written Opinion dated Sep. 21, 2012 for PCT Appl. No. PCT/US2012/025404. cited by applicant

Japanese Patent Office, Notification of Reasons for Refusal, JP Appl. No. 2013-554596, Dec. 8, 2015. cited by applicant

Chinese Office Action and Search Report for CN Appl. No. 201280014363, Mar. 23, 2015. cited by applicant

Chinese Office Action for CN Appl. No. 201280014363, Jan. 5, 2016. cited by applicant

Chinese Office Action for CN Appl. No. 201680045602.6 dated Jun. 5, 2019. cited by applicant

First Examination Report for Indian Appl. No. 6566/CHENP/2013 dated Aug. 29, 2019. cited by applicant

Beneq Biocompatible Coatings Webpage. cited by applicant

Cargille Laboratories, Inc. Material Safety Data Sheet—Cargille Optical Gel Code 0607, Jun. 3, 2005. cited by applicant

Depth of Field, OPMI Application Tip #2, Informed for Medical Professionals in Neuro, ENT and Spine, 2nd Issue, Oct. 2006, Published by Carl Zeiss Surgical GmbH, Germany. cited by applicant

Jaxton, et al., An Experimental Investigation on the Development of Hydrogels for Optical Applications, Polymer Testing, 2003, 22(4):371-374, English Abstract. cited by applicant

Kopp, et al., Chapter 9, Optical Principles of the Endoscope, Hysteroscopy: Visual Perspectives of Uterine Anatomy, Physiology & Pathology, 3rd Edition, Lippincott Williams & Wilkins, 2007, 19 pages. cited by applicant

Maquet Training Manual, Vasoview 6 Endoscopic Vessel Harvesting System, Cardiovascular, Copyright Maquet Cardiovascular LLC, Oct. 2008. cited by applicant

Oil Immersion, From Wikipedia, <http://en.wikipedia.org/wiki/Oil.sub.--immersion>, Printed Sep. 7, 2010. cited by applicant

Olympus Colonoscopes Outpatient Doctor Surgery Center, <http://outpatientsurgicare.com/index.PHP?Facilities:Technologies:Olympus.sub.--Colonoscopes&print>, Printed Oct. 26, 2010. cited by applicant

Olympus Disposal Distal Attachment Product Data Sheet. cited by applicant

Olympus Evis Exera Colonovideoscope/Sigmoidovideoscope, Olympus CF Type Q1601JUS, Today's Most Versatile Choice for Colonoscopy, Product Data Sheet. cited by applicant

Olympus Technologies Evis Exera II, Learn About Wide—Angle, <http://www.olympusamerica.com/msg.sub.--section/msg.sub.--endoscopy.sub.--technology.asp>, Copyright 2010 Olympus America Inc. cited by applicant

Olympus Technologies Evis Exera II, Learn About Close Focus, <http://www.olympusamerica.com/msg.sub.--section/msg.sub.--endoscopy.sub.--technology.asp>, Copyright 2010 Olympus America Inc. cited by applicant

Olympus NA-11J-KB Product Data Sheet. cited by applicant

Optical Gels for Fiber-Optic Connectors and Splices—A Tutorial, Nye Optical Products, 6 pages.

cited by applicant
Paxton, et al., An Experimental Investigation on the Development of Hydrogels for Optical Applications, Polymer Testing, 2003, 22(4):371-374, English Abstract. cited by applicant
Sigma-Aldrich Poly(2-hydroxyethyl methacrylate) Product Data Sheet,
[http://www.sigmaaldrich.com/catalog/Product Detail](http://www.sigmaaldrich.com/catalog/Product%20Detail), Copyright 2010 Sigma-Aldrich Co. cited by applicant
Sigma-Aldrich Poly(ethylene glycol) Product Data Sheet,
[http://www.sigmaaldrich.com/catalog/Product Detail](http://www.sigmaaldrich.com/catalog/Product%20Detail), Copyright 2010 Sigma-Aldrich Co. cited by applicant
Sigma-Aldrich Poly(vinyl alcohol) Product Data Sheet,
[http://www.sigmaaldrich.com/catalog/Product Detail](http://www.sigmaaldrich.com/catalog/Product%20Detail), Copyright 2010 Sigma-Aldrich Co. cited by applicant
Sigma-Aldrich Methacrylic acid Product Data Sheet, [http://www.sigmaaldrich.com/catalog/Product Detail](http://www.sigmaaldrich.com/catalog/Product%20Detail), Printed Sep. 3, 2010. cited by applicant
SmartGel Nye Nyogel OCK-451LPH Product Data Sheet, Nye Optical Products. cited by applicant
Smeds, et al., Photocrosslinkable Polysaccharides for in situ Hydrogel Formation, Journal of Biomedical Materials Research, 2001, 54:115-121. cited by applicant
Stadler, Transparent conducting oxides—An up-to-date overview, Materials 5.4:661-683, 2012. cited by applicant
The Basics of Silicon Chemistry, Basic Silicon Production and Siloxane Polymerization, <http://www.dowcorning.com/content/sitech/sitechbasics/siloxane.sub.--poly-merization.asp>, Copyright 2000-2010 Dow Corning Corporation. cited by applicant
Uw Eye Research Institute, Newsletter, Point of View, Summer 2009,
<http://vision.wisc.edu/news.sub.--sum09.html>, Printed Feb. 5, 2010. cited by applicant
Vinyl Sustainability Forum 2014, Title: Benefits of PVC, Date retrieved: Mar. 7, 2014 from website: <http://www.pvc.org/en/p/benefits-of-pvc>, pp. 1-4. cited by applicant
Zeng, et al., An Endoscope Utilizing Tunable-Focus Microlenses Actuated through Infrared Light, Solid-State Sensors, Actuators and Microsystems Conference, 2009, Transducers 2009, International, Issue 21-25, pp. 1214-1217, Abstract Only. cited by applicant
Zeng, et al., Tunable Liquid Microlens Actuated by Infrared Light-Responsive Hydrogel, Applied Physics Letters, 2008, 93:151101-1-151101-3. cited by applicant
Republic of China Patent Office; Office Action; Chinese Patent Application No. 201680054885.0; Jun. 3, 2020. cited by applicant
International Search Report and Written Opinion dated Dec. 16, 2020 for PCT Appl. No. PCT/US2020/65429. cited by applicant
International Search Report and Written Opinion dated Dec. 16, 2020 for PCT Appl. No. PCT/US2020/65424. cited by applicant
International Search Report and Written Opinion dated Dec. 16, 2020 for PCT Appl. No. PCT/US2020/65427. cited by applicant
International Preliminary Report on Patentability; PCT/US2021/026528; GI Scientific, LLC; Oct. 20, 2022. cited by applicant

Primary Examiner: Fairchild; Aaron B

Attorney, Agent or Firm: Farber LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/572,850, filed Jan. 11, 2022, which is a continuation of U.S. patent application Ser. No. 16/897,098 filed Jun. 9, 2020, issued as U.S. Pat. No. 11,253,137 on Feb. 22, 2022, which is a continuation of U.S. patent application Ser. No. 15/746,196 filed Jan. 19, 2018, issued as U.S. Pat. No. 10,856,724 on Dec. 8, 2020, which claims priority to International Patent Application Serial No. PCT/US16/43371, filed Jul. 21, 2016, which claims the benefit of U.S. Provisional Application No. 62/195,291, filed Jul. 21, 2015, all of which are hereby incorporated by reference for all purposes as if copied and pasted herein.

FIELD

(1) The present disclosure relates to a medical device, and more particularly to a device for covering and extending a working end portion of an optical imaging endoscope. The device provides a flexible working channel extension in communication with the working channel of the endoscope so that instruments can exit out of the working end portion at various angles, while also protecting the working end portion from ingress of bacteria, tissue, fluid, and other debris which could lead to infection and decreased performance of the scope.

BACKGROUND

(2) Recent advances in optical imaging technology have allowed many medical procedures to be performed today in a minimally invasive manner. The evolution of the more sophisticated, flexible scope with advanced visual capabilities has allowed access to regions deep within the human body that could only be achieved before with invasive surgical intervention. This modern day convenience has resulted in an increase in the demand for, as well as the number of, endoscopic, laparoscopic, arthroscopic, ophthalmoscopic, borescopic, or other remote imaging visualization procedures performed every year in the U.S. While these procedures are relatively safe, they are not without risks.

(3) Endoscopy, for instance, is a procedure in which a lighted visualization device called an endoscope is inserted into the patient's body to look inside a body cavity or organ for the purpose of examination, diagnosis or treatment. The endoscope may be inserted through a small incision or through a natural opening of the patient. In a bronchoscopy, the endoscope is inserted through the mouth, while in a sigmoidoscopy, the endoscope is inserted through the rectum. Unlike most other medical imaging devices, endoscopes are inserted directly into the organ.

(4) Today, most endoscopes are reused. This means that, after an endoscopy, the endoscope goes through a cleaning, disinfecting or sterilizing, and reprocessing procedure to be introduced back into the field for use in another endoscopy on another patient. In some cases, the endoscope is reused several times a day on several different patients.

(5) While the cleaning, disinfecting and reprocessing procedure is a rigorous one, there is no guarantee that the endoscopes will be absolutely free and clear of any form of contamination. Modern day endoscopes have sophisticated and complex optical visualization components inside very small and flexible tubular bodies, features that enable these scopes to be as effective as they are in diagnosing or treating patients. However, the trade off for these amenities is that they are difficult to clean because of their small size, and numerous components. These scopes are introduced deep into areas of the body which expose the surfaces of these scopes to elements that could get trapped within the scope or adhere to the surface, such as body fluids, blood, and even tissue, increasing the risk of infection with each repeated use.

(6) Endoscopes used in the gastrointestinal tract, especially specialty endoscopes also known as duodenoscopes with a side-viewing capability, have an added complexity in that they are in a bacteria rich environment. Typical duodenoscopes have internal moving components like an elevator with hinges attached to a cable for actuation. The elevator is used to deflect and therefore change the direction of instruments passed down the scope's working channel. This elevator is beneficial in that it can allow the user to change the direction of a wire or a catheter to direct the

wire or catheter into a specific opening, so that one or more instruments can be turned to enter a particular body lumen. However, given the size, location and movement of the elevator during use, the elevator creates cleaning issues, including the risk that bacteria finds its way into the elevator's hinges and other hard to clean locations on the scope. This provides an opportunity for bacteria to colonize and become drug resistant, creating the risk of significant illness and even death for a patient. This infection risk is also present in the cable mechanisms that are used to move the elevator mechanism back and forth and in other aspects of current scope designs. Moreover, in addition to the health risks posed by bacterial contamination, the accumulation of fluid, debris, bacteria, particulates, and other unwanted matter in these hard to clean areas of the scope also impact performance, shortening the useful life of these reusable scopes.

(7) Accordingly, it is desirable to provide devices which serve as convenient accessories for currently existing endoscopes to reduce the risk of contamination and infection, while also improving the performance of the endoscope. It is particularly desirable to provide an accessory for a duodenoscope that allows the user simultaneously to protect the working end from bacterial contamination and also enable instruments to exit out of the working end of the scope at different angles with ease.

SUMMARY

(8) The present disclosure provides a coupler device for covering and sealing a portion of the working end of a side viewing endoscope, with a flexible and tubular working channel extension that extends the working channel of the scope and can be angularly adjustable. The coupler device protects the scope and its components, particularly the scope elevator, to reduce the risk of debris, fluid and other matter ending up in the elevator and behind the elevator and the working channel, potentially causing infection risk and, in some embodiments, the device has its own way to articulate instruments, eliminating the need to have a scope with an elevator. The device may be single use disposable or reusable.

(9) The coupler device may be provided as a single-use disposable accessory to an endoscope that provides the user with the ability to change the angle of exit of a device being advanced out of the working channel of an endoscope, without exposing the distal end of the scope to bacteria, debris, fluid and particulate matter. In some embodiments, the device attaches to the end of the endoscope and covers the working channel of the endoscope with a working channel extension in the coupler device, allowing an instrument to be passed down the working channel of the endoscope and into the working channel extension of the coupler device. The working channel extension can provide a seal against the scope working channel, so instruments can be passed back and forth through the scope working channel and out the working channel extension of the coupler device without fluid and bacteria entering areas outside of the scope working channel. This seal is accomplished, in some embodiments, through an extension of the device working channel into the scope working channel, through a gasket on the end of the working channel extension, by way of a temporary glue, through pressure and the seal of the overall device against the distal end of the scope, through the selection of elastic and elastomeric materials, and other suitable and alternative means.

(10) The working channel extension of the coupler device can be made of one or more materials with elastic properties. The materials can include biocompatible material(s) when the device is intended for medical applications, which may include, without limitation, elastic and elastomeric materials, as well as combinations of rigid and flexible materials, including silicone joined to polycarbonate and other materials joined to a biocompatible metal.

(11) In some embodiments, the working channel extension of the coupler device may include an elastic biocompatible material that reduces the friction involving in passing devices through the working channel extension, which is joined to a biocompatible metal, such as a coil spring, an additional elastic material that is joined to the biocompatible metal, to improve flexibility, reduce kinking and aid in sealing the working channel of the device against the endoscope's working channel.

(12) In some embodiments, the device allows the user to articulate the working channel of the device in the direction preferred by the user of the endoscope, so that a wire, catheter or other instrument being advanced down the working channel of the endoscope can direct the wire or catheter or other instrument in a preferred direction different than the angle at which the instrument would exit the endo scope if the coupler device was not in place or if an elevator in the scope is not used. This redirection of an instrument has the benefit of assisting with the navigation of the device, while not allowing fluid, debris, particulate matter, bacteria and other unwanted elements to enter hard to clean areas of the endo scope, especially at the distal end of the endoscope.

(13) The benefits of the invention include allowing the physician to change the angle of exit, so that one or more devices can be turned to enter a particular body lumen, such as a biliary duct or pancreatic duct, or other hard to reach area, including in non-medical procedures, while sealing the distal end of the scope to prevent infection and the intrusion of debris and particulate matter into interior elements of the scope that are hard to reach to effectively clean.

(14) In some embodiments, the device may be formed of an optically clear material that covers the end of the endoscope and seals the end of the endoscope, allowing visualization of the endoscope's camera without obscuring the view by the device. The optically clear material may also cover the endoscope's light guide to allow the light projected by the endoscope to illuminate the field of view of the endo scope. In some embodiments, the optically clear material may include navigation markers to orient the user when visualizing tissue, such as markers to identify the relative position of the scope as the user visualizes the tissue through the optically clear material.

(15) In embodiments, the optically clear material may also include other markers to guide the user with confirmation of the accurate placement of the optically clear material over the endoscope's camera and, if applicable, over the endoscope's light guide.

(16) In some embodiments, the device may articulate instruments through the device through a cable in a sealed sheath that is attached to the flexible working channel extension in the coupler device, allowing the user to advance and retract the cable to move the working channel extension backward and forward to change the angle of exit from the flexible working channel in order to direct an instrument to a desired direction.

(17) In some embodiments, the device may have multiple cables so the angle of exit can be articulated in multiple directions, including in different quadrants, unlike with the current endoscope elevators, which can only deflect and therefore redirect an instrument in a single axis due to the limited travel of endoscope elevators, which can only be raised or lowered, but not moved from side to side or articulated into other quadrants. In some embodiments, the cable(s) may be attached directly to the working channel extension or to other devices that can be articulated and cause the working channel extension to change its angle of exit, including, for example, a dowel underneath the working channel extension, but encased in the device that can be advanced forward and backward to move the working channel extension as the cable is advanced and retracted. In some embodiments, the articulation ability of the coupler device may be created with an elevator embedded in the coupler device, which is disposable and therefore thrown away after the procedure.

(18) The articulation ability of the coupler device may also take place with elements that do not involve cables, including for example, piezo electric materials, micro motors, organic semiconductors, and electrically activated polymers. In some embodiments, the articulation ability of the coupler device may also take place with the transfer of force to the working channel extension or an embedded elevator through interlocking connectors that transfer force, wires that twist, slidable sheaths, and memory metals that change shape through the transfer of temperature. In some embodiments, the device includes a power connector or motors to deliver energy, including electromagnetic energy, to the device to cause a transfer in force to change the angle of exit from the coupler device as an instrument is passed through the device, or in advance of passing an instrument through the device. This transfer of force can include causing the device to rotate as

it exits the working channel extension. The device may be navigated and articulated by the user directly, or as part of a robotic system in which the user's input is translated through the system through various means, including cables, power connectors, motors, electromagnetic energy, slideable sheaths, haptics, computer-guided and directed input, and other means to direct and guide the device to its intended location, including to specific diagnosis and treatment objectives in a patient, or in non-medical applications, to a desired remote location.

(19) In some embodiments, the device may be integrated into a scope and configured to be detachable and reusable for separate cleaning, including manual cleaning, in an autoclave, an ETO sterilizer, gamma sterilizer, and other sterilization methods.

(20) The articulation aspect of the coupler device may include a locking feature or capability to affix the angle of exit in the working channel extension at a specific angle. In some embodiments, the specific angle of exit may be aimed at a specific point in the gastrointestinal tract, such as a biliary or pancreatic duct, or the angle of exit may be affixed so that a wire or other instrument inside the working channel temporarily cannot be advanced, locking the instrument in position temporarily to aid in the exchange of instruments or to improve navigation of the instrument temporarily.

(21) The device may include a disposable or reusable control mechanism that attaches to the endoscope to articulate the distal end of the coupler device to change the angle of exit from the working channel extension of the coupler device. In some embodiments, this control mechanism may also lock the angle of exit of the working channel extension or the working channel extension may be locked through elements in the endoscope itself, such as the elements that articulate the endoscope's elevator.

(22) In some embodiments, the coupler device may cover the entire distal end of the endoscope, or may just cover hard to clean areas. In some embodiments, the coupler device may cover the distal end of the endoscope, or a portion thereof, or it may include a sheath attached to the coupler device which covers the entirety of the scope that is exposed to fluid, debris, particulate matter, bacteria and other unwanted elements.

(23) In some embodiments, the device includes an anti-infective material. In another exemplary embodiment, the device includes an anti-infective coating. In still another embodiment, the device includes a coating that is hydrophobic. In yet another embodiment, the device is superhydrophobic. In even still another embodiment, the device is anti-infective and hydrophobic. Further yet in another embodiment, the device is anti-infective and superhydrophobic. In further still another exemplary embodiment, anti-inflammatory coatings are incorporated into the device.

(24) The device may include a silver ion coating and a silver hydrogel applied, infused or made part of the device in the area that covers or goes around the scope elevators. The device may also include a valve or other element at the distal end of the catheter channel and may in embodiments have a valve in the working channel extension to prevent having fluid and debris traveling from the lumen back into the scope working channel.

(25) The device may include an electrical wire or other power transmission point to enable the creation of an electrical field across a silver ion coating to improve the activity of the silver ion coating or other coating to prevent infection.

(26) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosure. Additional features of the disclosure will be set forth in part in the description which follows or may be learned by practice of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments of the disclosure and together with the description, serve to explain the principles of the disclosure.
- (2) FIGS. **1A AND 1B** are isometric views of an exemplary embodiment of the coupler device of the present disclosure in use with a duodenum scope.
- (3) FIGS. **2A AND 2B** show partial cutaway views of the coupler device and a duodenum scope of FIGS. **1A AND 1B**, respectively.
- (4) FIG. **3** shows another cutaway view of the coupler device and a duodenum scope of FIGS. **1A AND 1B**.
- (5) FIG. **4** shows still another cutaway view of the coupler device and a duodenum scope of FIGS. **1A AND 1B**.
- (6) FIG. **5** is a cutaway side view of the coupler device and a duodenum scope of FIGS. **1A AND 1B** in a first position.
- (7) FIG. **6** is a cutaway side view of the coupler device and a duodenum scope of FIGS. **1A AND 1B** in a second position.
- (8) FIG. **7** is a cutaway side view of the coupler device and a duodenum scope of FIGS. **1A AND 1B** in a third position.
- (9) FIG. **8** is an enlarged side view of the working channel extension with membrane of the coupler device of FIGS. **1A AND 1B**.
- (10) FIG. **9** is a top-down view of the coupler device of FIGS. **1A AND 1B**.
- (11) FIG. **10** is a cutaway view of another exemplary embodiment of a coupler device of the present disclosure.
- (12) FIG. **11** is a cutaway side view of the coupler device of FIG. **10**.
- (13) FIG. **12** is a cutaway side view of the coupler device of FIG. **10** in use with a duodenum scope.
- (14) FIG. **13** is an enlarged side view of an exemplary embodiment of a working channel extension of the present disclosure.
- (15) FIG. **14** is another enlarged side view of the working channel extension of FIG. **13**.
- (16) FIG. **15A** is a perspective view of the working channel extension of FIG. **13** and FIG. **15B** shows the working channel extension of FIG. **15A** in use with an instrument.
- (17) FIG. **16** is a perspective top-down view of the coupler device of FIG. **1A** with a locking feature.
- (18) FIG. **17** is a perspective view of another exemplary embodiment of a working channel extension of the present disclosure.

DETAILED DESCRIPTION

- (19) Turning now to the drawings, FIGS. **1A** and **1B** illustrate an exemplary embodiment of a coupler device **10** of the present disclosure. The coupler device **10** serves as an accessory component for currently existing endoscopes. The device seals and covers infection prone areas of the scope to prevent ingress of debris, fluid, or other unwanted matter that could lead to bacterial contamination and decreased performance of the scope. In addition, the coupler device **10** provides a flexible working channel for instruments to be inserted into the scope. The flexible working channel can be angularly adjustable with ease. As shown, the coupler device **10** may be used with a duodenum scope **40** or other side-viewing scope instrument. It is understood, of course, that the coupler device **10** may be adapted for use with end viewing scopes as well. In addition, the coupler device **10** of the present disclosure can be used with all types of scopes for different medical applications. The duodenum scope **40** shown here is merely for illustrative purposes.
- (20) As FIGS. **1A** and **1B** illustrate, the coupler device **10** may comprise a main body **12**, proximal end **14** and distal end **16**, lower surface **18** and upper surface **20**. The proximal end **14** attaches onto

a working end of a duodenum scope **40**, extending the working end portion of the scope **40**. The upper surface **20** may include a lens and light guide **24** and a scope washer opening **28**, which is used to push fluid across the scope camera to wash debris off the camera and is also used to push air across the camera to dry the camera and insufflate the patient's gastrointestinal tract. In addition, the upper surface **20** includes a flexible working channel region **30** that includes a flexible working channel extension **34** that is surrounded by a flexible membrane **38**. This flexible membrane **38** serves as a protective hood or covering for the working end of the coupler device **10**, providing for flexible articulation while sealing out debris, fluid, bacteria or other unwanted matter.

(21) As shown in FIGS. 2A and 2B, the duodenum scope **40** may comprise a light guide **44**, lens **46** and washer opening **48**. The coupler device **10** cooperates with each of these components of the scope **40** to provide a fully functioning scope. The coupler device **10** does not interfere with the scope's ability to emit a clear image, but instead reduces the risk of contamination with each use. This benefit is achieved by providing a coupler device **10** which attaches to the working end components of the scope **40**, and seals around the working end.

(22) As further shown in FIGS. 1A, 1B 2A, 2B, 3 and 4, the coupler device **10** provides an extension of the scope's working channel **42**. The working channel extension **34** of the coupler device **10** in FIG. 1 is flexible and may contact the scope's working channel **42** by a sealed connection, as shown in FIG. 4, at the proximal end **34a** of the working channel extension. The distal end **34b** of the working channel extension **34** serves as an exit portal for instruments to pass through the scope **40** to reach different areas of the body.

(23) Additionally, the coupler device **10** provides a further seal around the elevator **50** of the scope. Because the coupler device **10** seals the elevator **40**, risk of debris influx, fluids, bacteria and other matter build up behind the elevator and working channel is reduced significantly. This influx of debris, bacteria and other matter is believed to be the reason for drug resistant infections with current scopes today. While preventing influx, the coupler device **10** advantageously maintains flexibility to move the working channel extension **34**.

(24) In use, the scope's working channel extension **34** permits passage of instruments down the scope working channel **42** and through and out the working channel extension **34** of the device **40** for assessment and treatment of tissue and other matter. Such instruments may include cannula, catheters, stents and stent delivery systems, papillotomes, wires, other imaging devices including mini-scopes, baskets, snares and other devices for use with a scope in a lumen. This working channel extension **34** is flexible enough that the elevator **50** of the scope **40** can raise and lower the working channel extension **34** so that instruments can be advanced down and out of the working channel extension distal end (or exit portal) **34b** of the scope **40** at various angles, or be raised and lowered by a cable or other means to articulate the working channel extension **34**.

(25) As FIGS. 5 to 7 illustrate, in use when the elevator **50** of the scope **40** is actuated, the flexible working channel extension **34** of the coupler device moves or adjusts to this actuation, along the direction A-A In FIG. 5, the elevator **50** is raised slightly, creating a hinged ramp or shoulder that pushes the working channel extension **34** a corresponding angle and shifts the exit portal or distal end **34b** of the working channel extension to the left. In FIG. 6 the elevator is raised higher than in FIG. 5, such that the distal end **34b** of working channel extension **34** is likewise shifted further to the left in comparison to FIG. 5, while FIG. 7 shows the elevator **50** raised even higher and the distal end **34b** of working channel extension **34** moved to the left even further in comparison to FIGS. 5 and 6.

(26) As FIG. 8 shows, the ability of the distal end **34b** of working channel extension **34** to shift along the width of the working channel region **30** of the coupler device **10** is in part due to the fact that the distal end **34b** is itself attached to a flexible membrane **38**. This flexible membrane **38** comprises a plurality of loose folds or creases, allowing the excess material to stretch and bend as the elevator actuation forces the working channel extension to bend and shift in response. In addition, the flexible membrane **38** acts as a protective cover or hood for the working channel

region **38**, preventing the ingress of fluids, debris, or other unwanted matter from getting inside the scope **40** and causing a bacterial contamination or the infusion of other unwanted fluid, debris or particulate matter.

(27) It is contemplated that the coupler device **10** of the present disclosure may be configured for single, disposable use, or it may be configured for reuse. The coupler device **10** may be made of any biocompatible material, such as for example, silicone or another elastic or polymeric material. In addition, the material may be transparent. As shown in FIG. **9**, the coupler device **10** may be formed of a transparent material to provide a transparent covering of the scope camera and light source, thereby allowing unhindered performance of the scope **40**.

(28) FIGS. **10** to **12** show another exemplary embodiment of a coupler device **10** of the present disclosure. In this embodiment, the coupler device **10** is adapted for use with scopes that are actuated by cable and eliminates the need for the elevator component. As illustrated, the coupler device **10** maintains the same structural features as previously described, but now includes a further disposable external sheath **60** that can receive an interior actuating cable **54** of the scope. This cable **54** can be detached from the elevator and reattached to the flexible working channel extension **34** of the coupler device **10**. The elevator is no longer needed in this embodiment, as actuation of the cable effects movement of the working channel extension **34**. The external sheath **60** may be configured to attach directly to the scope **40**, such as by winding around the outside of the scope or by a friction fit connection. In embodiments, multiple cables may be included in one or more sheaths to provide for articulation in other quadrants than the single axis articulation with elevators in current duodenoscopes.

(29) In other embodiments, the coupler device **10** may also include a closable port (i.e., self-sealing) that allows for the injection of anti-adhesion, anti-bacterial, anti-inflammatory or other drug or infusible matter that prevents the adherence or colonization of bacteria on the scope. An applicator may be provided that is integrated into the coupler device **10** with a port for delivery of the infusible matter. Alternatively, the applicator may be separate from the coupler device **10** and applied to the distal end of the scope **40**. The infusible matter may include forms of silver, including in a gel or other solution, platinum, copper, other anti-adhesion, anti-bacterial, anti-inflammatory or other drug or infusible matter that is compatible with the scope and coupler device materials and biocompatible for patient use.

(30) In one exemplary embodiment, the device includes an anti-infective material. In another exemplary embodiment, the device includes an anti-infective coating. In still another embodiment, the device includes a coating that is hydrophobic. In yet another embodiment, the device is superhydrophobic. In even still another embodiment, the device is anti-infective and hydrophobic. Further yet in another embodiment, the device is anti-infective and superhydrophobic. In further still another exemplary embodiment, anti inflammatory coatings are incorporated into the device.

(31) In one exemplary embodiment, the device **10** may include a silver ion coating. In another embodiment, the device **10** may have a silver hydrogel applied, infused, or made part of the device **10** in the area that covers or goes around the scope elevators. In addition to silver having antimicrobial properties, silver can also conduct electricity. Thus, in still another embodiment, the device **10** may include an electrical wire or other power transmission point to enable the creation of an electric field across the silver ion coating to improve the ability of the silver ion coating to prevent infection. In some embodiments, the electrical wire or other power transmission point may also apply to other antimicrobial and conductive materials, including platinum and copper.

(32) FIGS. **13** and **14** show another embodiment of the working channel extension **134** of the present disclosure. As contemplated, the working channel extensions may comprise a combination of different materials. For example, as shown in FIG. **13**, the working channel extension **134** may be formed of multiple elastic materials joined to a biocompatible metal. In some embodiments, one of the elastic materials may be PTFE and another elastic material may be a biocompatible elastic material that covers the biocompatible metal. In the example of FIG. **13**, the working channel

extension **134** may comprise an inner elastic material **110** and an outer elastic material. The outside of the working channel extension **134** may embody, the biocompatible metal may be encapsulated by one or more of the elastic materials.

(33) In FIG. **14**, the outer biocompatible elastic material **120** is formed to create a gasket **122** to seal the proximal end of the working channel extension against **134** the working channel of an endoscope, creating a seal to prevent the intrusion of unwanted bacteria, biomatter and other material into this sealed area.

(34) In FIG. **15A**, a working channel extension **134** is shown with an adjustable angle of exit **8** for locking an instrument **100** in place. In this embodiment, when the angle of exit **8** is adjusted, it creates compressive force in the working channel **134**, locking an instrument **100** in place, as shown in FIG. **15B**. This can be used to fixate an instrument while a wire is advanced through the instrument, or to fixate a wire, while a second instrument is exchanged over the wire.

(35) In FIG. **16**, an alternative embodiment is shown for locking an instrument **100** in place. In this embodiment, the working channel extension **134** is raised to a point in which the instrument **100** in the working channel extension **134** is compressed against a lock **80** on the device **10**, causing a change in the angle of exit of the working channel extension **134** and locking the instrument **100** in a fixated place in the working channel extension **134**.

(36) In FIG. **17**, an alternative embodiment of the working channel extension **234** is shown with a flange **268** for attaching the working channel extension to the membrane material **38** that is part of the device **10**.

(37) Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the embodiment disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the embodiment being indicated by the following claims.

Claims

1. A device for use with an endoscope having an elongate shaft with an internal working channel, the device comprising: a sheath; a main body comprising a visualization section for allowing viewing of tissue by the endoscope and a proximal end configured for attachment to a distal end portion of the endoscope; a passage within the main body having an exit portal and a proximal end configured for alignment with the working channel of the endoscope; a flexible membrane in contact with, and coupled to, an outer surface of the main body and at least partially surrounding the exit portal; and wherein the device is configured such that an instrument passing through the passage is capable of angular adjustment relative to the main body by actuation of the endoscope.
 2. The device of claim 1, wherein the passage comprising a working channel extension within the main body.
 3. The device of claim 2, wherein the working channel extension is flexible and capable of angular adjustment by actuation of the endoscope.
 4. The device of claim 1, wherein the proximal end of the passage is configured for insertion into the working channel of the endoscope.
 5. The device of claim 1, wherein the main body of the device is configured to accommodate therein an elevator for angular adjustment of the instrument.
 6. The device of claim 1, further comprising a mechanism for articulating an instrument passing through the passage.
 7. The device of claim 1, wherein the main body comprises an elastic or elastomeric material.
 8. The device of claim 1, wherein the endoscope is a side-viewing scope.
 9. The device of claim 1, wherein the endoscope includes an elevator for angular adjustment of the instrument.
-

